

Artificial Intelligence (AI) – Viva Notes

1. Introduction to Artificial Intelligence

Artificial Intelligence is a branch of computer science that develops systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, perception, language understanding, and decision making.

2. Types of Artificial Intelligence

Narrow AI: AI designed to perform a specific task, such as speech recognition or recommendation systems.

General AI: A theoretical form of AI that would perform a wide range of intellectual tasks at a human-like level.

Superintelligent AI: A hypothetical form of AI whose capabilities would exceed human intelligence in many areas.

3. Machine Learning

Machine Learning (ML) is a subfield of AI in which computer systems learn patterns from data and use those patterns to make predictions or decisions without being explicitly programmed for every case.

4. Types of Machine Learning

- **Supervised Learning:** The model learns from labeled training data. Examples include classification and regression.
- **Unsupervised Learning:** The model finds patterns in data without labeled outputs. Examples include clustering and dimensionality reduction.
- **Reinforcement Learning:** An agent learns by interacting with an environment and receiving rewards or penalties.

5. Deep Learning

Deep Learning is a part of machine learning that uses neural networks with multiple layers. It is widely used in image recognition, speech processing, natural language processing, and other complex tasks.

6. Neural Networks

An artificial neural network is a computational model inspired by the structure of biological neural systems. It commonly contains an input layer, one or more hidden layers, and an output layer.

7. Natural Language Processing

Natural Language Processing (NLP) is a field of AI that enables computers to process, understand, and generate human language. Applications include chatbots, translation, sentiment analysis, text classification, and question answering.

8. Generative AI

Generative AI refers to AI systems that can create new content such as text, images, audio, video, or code based on learned patterns.

9. Large Language Models

A Large Language Model (LLM) is a machine learning model trained on large amounts of text to understand and generate natural language. LLMs can be used for question answering, summarization, coding assistance, and conversational applications.

10. Retrieval-Augmented Generation (RAG)

RAG combines information retrieval with text generation. First, relevant information is retrieved from a knowledge source, such as a document database. The retrieved information is then supplied to a language model to generate a more context-aware response.

11. Embeddings

An embedding is a numerical representation of text or other data. Similar meanings tend to have similar vector representations. Embeddings are commonly used for semantic search and document retrieval.

12. Vector Database

A vector database stores vector representations and supports similarity searches. In a RAG system, it can store document chunks and their embeddings so that relevant information can be retrieved for a user query.

13. AI Viva Examiner Project

An AI Viva Examiner can use a student's study PDF as its knowledge source. The PDF text can be extracted, divided into chunks, converted into embeddings, and stored in a vector database. When a viva question is required, relevant context can be retrieved and provided to a local language model such as an Ollama-supported model.

14. Technologies Used in the Project

- **Streamlit:** Builds the web interface.
- **PyPDF:** Extracts text from PDF files.
- **Sentence Transformers:** Creates text embeddings.
- **ChromaDB:** Stores and retrieves vector embeddings.
- **Ollama:** Runs a local large language model.

15. Important AI Viva Questions

- 1 What is Artificial Intelligence?
- 2 What are the main goals of AI?
- 3 What is the difference between AI and Machine Learning?
- 4 What is supervised learning?
- 5 What is unsupervised learning?
- 6 What is reinforcement learning?
- 7 What is deep learning?
- 8 What is a neural network?
- 9 What is NLP?
- 10 What is Generative AI?
- 11 What is an LLM?
- 12 What are embeddings?
- 13 What is a vector database?
- 14 What is RAG?
- 15 Why is RAG useful in a document-based AI application?
- 16 What is the role of PyPDF in the AI Viva Examiner?
- 17 What is the role of Sentence Transformers?
- 18 Why is ChromaDB used?
- 19 What is the role of Ollama?
- 20 What is Streamlit?
- 21 How does an AI Viva Examiner generate a question from a PDF?
- 22 How can an AI system evaluate a student's answer?
- 23 What are some limitations of AI systems?